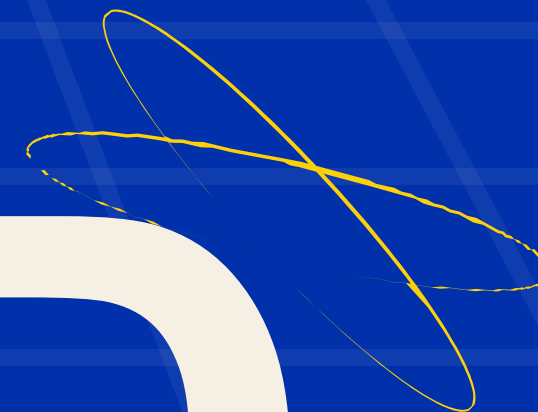




ROUND TABLE



— PROJECT PROPOSAL —

Presented by : The Keystrokes

October 4, 2026

THE

BREAKDOWN



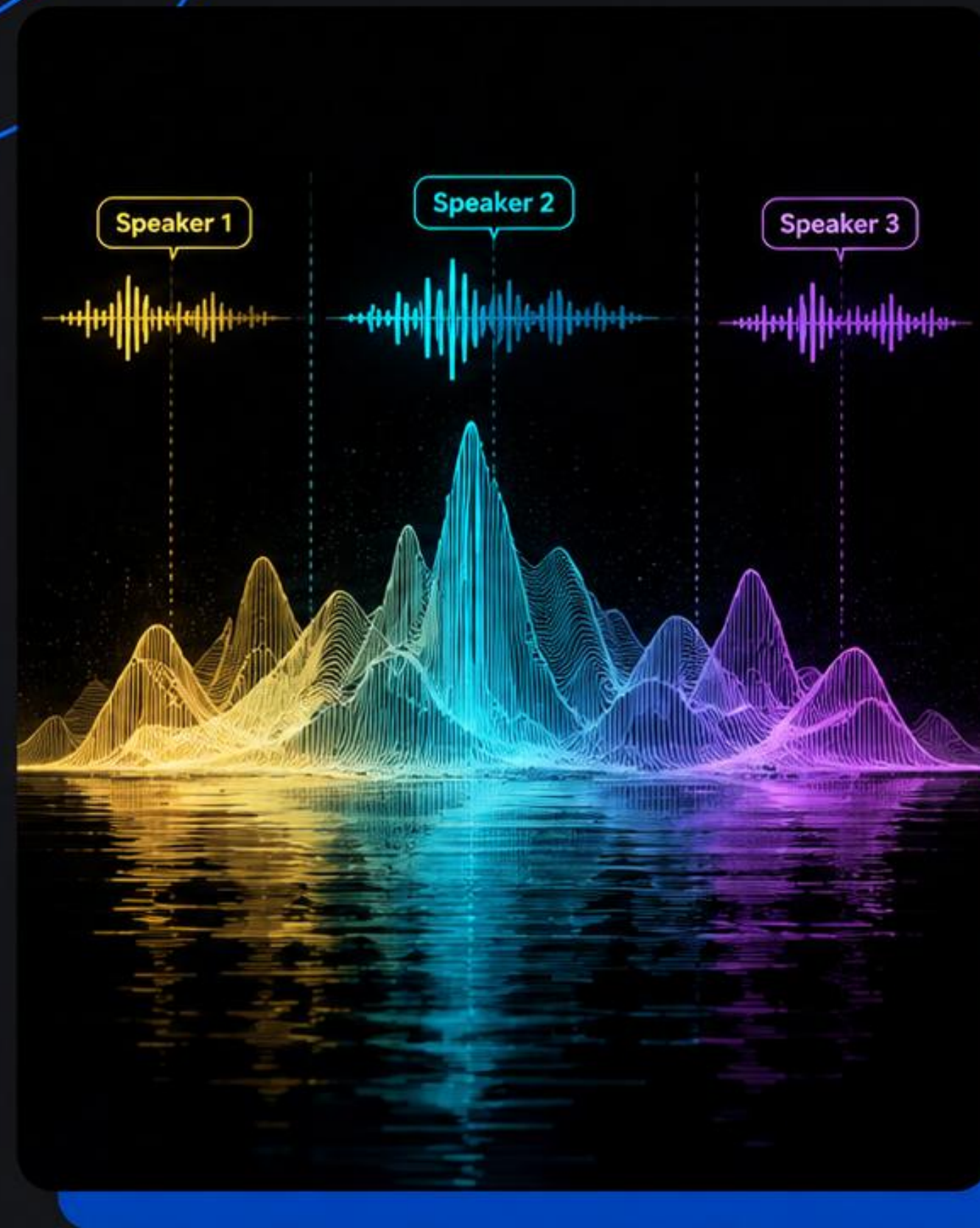
The Cocktail Party Problem

In group meetings, multiple people often speak at the same time, **overlap and interrupt**. Conventional single-stream captions merge all voices into one, making it hard to understand who said what and **losing speaker attribution**.



Cognitive Overload & Exclusion

Deaf or hard-of-hearing participants, and people in noisy environments, struggle when captions lag, merge multiple speakers, or fail to identify who is speaking. This makes real-time participation difficult and can lead to **missing context** and a less inclusive meeting experience.



WebRTC gives us the meeting layer. **The missing layer is low-latency, speaker-aware captions.**

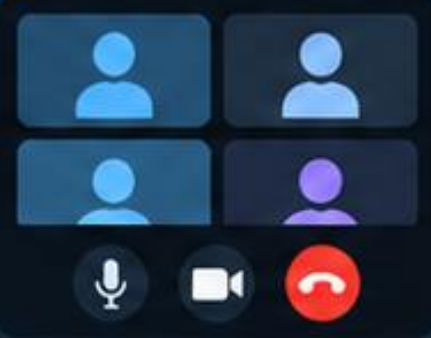
SOLUTION OVERVIEW

REAL-TIME CAPTIONS FOR SMOOTHER MEETINGS

We enable **live transcription** during video meetings so that everyone can follow the conversation in real-time, even if they miss words, speak in different accents, or cannot hear clearly.



01



Join a Meeting

Create or join a room using a unique link.

02



Capture Audio

The meeting audio is captured from the microphone in real-time.

03



Convert to Text

The audio is instantly converted into text while the meeting is in progress.

04



Live Captions

Accurate, real-time captions are shown on screen so everyone can follow the conversation.

KEY BENEFITS



Better Understanding

Helps users follow the conversation easily, even in noisy environments.



More Inclusive

Useful for people with hearing difficulties or language barriers.



Real-Time Support

Captions appear instantly as people speak, without disrupting the meeting.



Works in Browser

No extra setup required. Just join from any modern browser.

PIPELINE

REAL-TIME VIDEO MEETING ARCHITECTURE

Built with **Rust**, **Leptos** and **Axum**. Uses **WebRTC** for direct peer-to-peer audio/video and **WebSocket** for signaling for room coordination.



1. Camera / Mic (Acquire Media)

Browser captures audio & video using `getUserMedia()`



2. WebRTC (Peer Connection)

Create peer-to-peer connection and send/receive audio-video streams



3. WebSocket (Signaling)

Exchange offer/answer and ICE candidates through Axum server



4. Real-Time Video Meeting

Multiple participants connected in a room with audio/video streams

01

Media Capture (Browser)



Uses browser's `getUserMedia()` to access camera and microphone. The media stream is added to the `RTCPeerConnection` and sent to other peers.

```
navigator.mediaDevices.getUserMedia({
  video: true,
  audio: true
})
```

02

WebRTC (Peer-to-Peer)



Creates a separate `RTCPeerConnection` for each participant. Exchanges `offer/answer` and `ICE candidates` to establish direct peer-to-peer audio/video streams.

```
pc.createOffer()
pc.setRemoteDescription()
pc.addIceCandidate()
```

03

WebSocket Signaling (Axum Server)



A lightweight `Axum` server handles signaling. It manages rooms, forwards offer/answer and ICE candidates between peers. Current implementation supports up to **16 participants** per room (in-memory).

```
{
  "type": "offer",
  "room_id": "abc123",
  "target": "peer_id",
  "sdp": "..."
}
```



USER EXPERIENCE

REAL-TIME VIDEO MEETINGS BUILT ON WEBRTC

01



Dynamic Multi-Participant Grid

When multiple people join the same room, their video streams are shown in a responsive grid in real-time using WebRTC peer-to-peer connections. Users can see and hear each other with low latency, just like a modern video meeting app.

02



Room-Based Collaboration

Users can create or join a room using a unique room ID. People joining the room are instantly connected via WebSocket signaling and WebRTC. Join/leave events are handled in real-time, with support for multiple participants (up to 16 per room).

TECHNICAL OVERVIEW

REAL-TIME VIDEO MEETING APPLICATION
BUILT WITH RUST, LEPTOS AND AXUM



Browser (Client)



Camera & Mic
getUserMedia()



WebRTC
Peer Connection

WebSocket
(Signaling)



Rust Server (Axum)



Signaling Server
Room Management
Message Relay



In-Memory Rooms
(up to 16 participants)

WebSocket
(Signaling)



Other Participants



Peer A
WebRTC Stream
(Audio/Video)



Peer B
WebRTC Stream
(Audio/Video)



TECH STACK



Rust

Core language



Leptos

Frontend
(SSR + WASM)



Axum

WebSocket
Signaling Server



WebRTC

Peer-to-Peer
Audio/Video

KEY FEATURES (CURRENT IMPLEMENTATION)



Browser camera &
microphone access
getUserMedia()



WebRTC peer-to-peer
connections
Offer / Answer, ICE



Multiple participants
per room
(Max 16)



WebSocket signaling
Room join/leave
Message relay



Real-time audio/video
streams
Local + remote streams



In-memory room
management
Rooms stored in server
memory (not persisted)



THANK
YOU

